

Developing Multisource Datawarehouse for Hybrid Machine Learning Female Recommendations in E-commerce Platform

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Abstract—

Index Terms—E-commerce platforms increasingly rely on recommendation systems to improve user experience and increase sales. However, traditional approaches often fail to understand contextual factors such as regional preferences and temporal trends. This paper proposes a hybrid machine learning-based recommendation system that integrates customer behavior analysis, association rule mining, and clustering techniques to improve recommendation accuracy. In particular, the Apriori algorithm is used to find repeated purchase patterns, and K-Means clustering is used for client segmentation. In order to provide recommendations that are more relevant and timely, the suggested model takes into account contextual factors including geographic location and seasonal patterns. For instance, the suggested method successfully captures the large regional and temporal variations in demand for particular products, such as clothing that is culturally appropriate or things associated to Ramadan. Additionally, preference and sentiment analytics from customer reviews are used to improve the quality of recommendations. The hybrid and context-aware strategy works better than standard recommendation techniques, according to experimental evaluation using common metrics like accuracy, precision, recall, and F1-score. The findings emphasize how crucial it is to combine machine learning with contextual data in order to produce more effective and personalized e-commerce recommendations.

Keywords— E-commerce, Recommendation System, Machine Learning, K-means, Apriori, Regional Data

I. INTRODUCTION

E-commerce has expanded rapidly in recent decades, transforming the way customers discover and purchase products. As online catalogs continue to grow, users are exposed to an overwhelming number of alternatives, making recommendation systems essential for reducing information overload and improving personalization. Early collaborative filtering research demonstrated that user behavior can be leveraged to generate useful recommendations, laying the foundation for modern recommender systems used by large-scale platforms and online retail environments [1]. Recent studies also show that data-driven recommendation frameworks, sentiment-aware product analysis, and fashion-oriented clustering can improve recommendation quality in e-commerce settings [2], [3]. Despite

their success, traditional recommendation approaches still face important limitations. Collaborative filtering remains vulnerable to the cold-start problem for new users and items, while many systems do not fully capture contextual signals such as customer behavior, transaction patterns, review sentiment, or product appearance. Given the scale and heterogeneity of e-commerce data, data mining techniques are essential for extracting hidden patterns, customer behavior insights, and product relationships that cannot be captured through traditional recommendation approaches alone. To address these gaps, this project proposes a hybrid machine learning framework that combines data warehousing, exploratory analytics, sentiment analysis, image classification, churn prediction, and collaborative recommendation into a unified pipeline. In this research, we implement a Medallion-style data warehousing architecture using MySQL to manage high-volume e-commerce datasets. The Bronze layer stores raw data from sources such as Amazon product records, Women's Clothing E-Commerce Reviews, and Online Retail II transaction logs. The Silver layer applies structured SQL-based cleansing to improve data quality through the removal of cancellations, business-key deduplication, and the elimination of HTML and URL noise from textual reviews. The Gold layer organizes the cleansed data into a high-performance star schema containing fact tables and normalized dimension tables for time, location, product, and customer, producing a single analytical source of truth for downstream predictive modeling. Before modeling, the project performs exploratory data analysis and diagnostic profiling to uncover behavioral patterns. Recency, Frequency, and Monetary analysis is used to segment customers into groups such as Loyalists, At-Risk, and Hibernating. Market basket analysis is then applied through the Apriori algorithm to identify association rules and evaluate product bundling potential using support, confidence, and lift. In parallel, sentiment trends are examined to relate textual polarity to star ratings and reveal cases of hidden dissatisfaction in customer reviews. These diagnostics provide a deeper understanding of purchasing behavior and review dynamics before predictive modeling begins [2], [4]. The modeling framework is de-

signed to prevent data leakage through a strict 70/15/15 train-validation-test split with stratification and, where necessary, subject-level partitioning. All numeric features are scaled in a unified feature space so that every model receives consistent inputs, even when some algorithms are scale-invariant. Review information is aggregated at the product-entity level rather than retained as isolated review rows, which enables more realistic item profiles that combine sentiment and visual attributes. The churn prediction component uses a champion–challenger setup with Logistic Regression, Random Forest, XGBoost, SVM, and MLP classifiers, with SMOTE applied to address class imbalance [5]. The sentiment module fuses VADER and TextBlob scores after normalization, the image module uses a CNN to classify apparel into 14 categories, and the recommender combines collaborative filtering with content-based filtering using cosine similarity and content-derived item features. This hybrid design helps address sparsity and cold-start limitations while improving recommendation relevance [1]–[3]. Overall, this paper proposes a hybrid e-commerce recommender that integrates customer segmentation, association-rule mining, sentiment analysis, image-based classification, and collaborative filtering within a scalable warehouse-backed architecture. By combining multiple datasets and machine learning techniques in one framework, the proposed system aims to improve recommendation quality, support data-driven decision-making, and enhance sales performance in online retail [2], [3].

II. RELATED WORK

Several studies have explored customer behavior analysis and recommendation systems using the Amazon Product Reviews dataset and retail transaction data. Atre and Tapaswi [5] proposed a machine learning-based approach for predicting customer behavior using sentiment analysis techniques applied to customer reviews. Several studies have explored customer behavior analysis and recommendation systems using the Amazon Product Reviews dataset and retail transaction data. Atre and Tapaswi [5] proposed a machine learning-based approach for predicting customer behavior using sentiment analysis techniques applied to customer reviews. Preprocessing textual data, feature extraction, and classification using Naïve Bayes and Logistic Regression constituted their technique. With an accuracy of 96.62 percentage, the experimental findings demonstrated that Logistic Regression performed better. Nevertheless, the study’s usefulness in modern e-commerce environments is restricted because it only uses traditional classification models and ignores recommendation systems and hybrid approaches. Turki et al. [2] introduced a scalable, data-driven framework for enhancing e-commerce recommendation systems using the Amazon Product Reviews dataset. Their approach integrates collaborative filtering, content-based filtering, and sentiment analysis within a distributed computing environment using Apache Spark. The model demonstrated improved performance with metrics such as RMSE of 0.95, precision@k of 0.85, and recall of 0.82. Despite these improvements, the approach suffers from high computational

complexity and is limited in generalization due to its focus on specific product categories. Chen and Gunawan [6] proposed a recommendation system based on retail transaction data using a modified RFM model that incorporates additional features such as periodicity and customer engagement index. Their approach combines clustering techniques, including K-means and Ward’s method, with association rule mining to identify purchasing patterns and generate recommendations. The results showed improved segmentation and recommendation accuracy, particularly when using K-nearest neighbors for validation. This study highlights the importance of integrating behavioral analytics with transactional data for more effective recommendation systems. Yadav [4] developed an aspect-based sentiment analysis framework for product recommendation using customer reviews. The WordNet lexical database is used to extract important product qualities, and rule-based sentiment classification is used to assess each feature. Compared to standard rating-based methods, the system offers more detailed recommendations by calculating weighted sentiment ratings. Recommendation reliability may be impacted by the method’s difficulties when handling scant review data. A multi-clustering recommendation system for fashion retail was proposed by Bellini et al. [3], in which consumers and goods are categorized according to their characteristics and habits. Products are grouped according to characteristics like brand and category, while users are divided into groups according to purchasing patterns, frequency, and date of purchase. To enhance personalization and solve cold-start problems, the system integrates several recommendation techniques, such as comparable users, similar objects, and association-based recommendations. The outcomes of the experiment showed higher user engagement and purchase rates. Scalability is still a problem for large and dynamic datasets, and the model is highly dependent on data quality and clustering performance. In addition, deep learning techniques have been applied to enhance product understanding in recommendation systems. Kumar et al. [7] investigated clothing image classification using Convolutional Neural Networks on the Fashion-MNIST dataset. Their results showed that CNN-based models outperform traditional methods in accuracy and generalization, highlighting the effectiveness of deep learning for visual feature extraction in e-commerce applications.

III. METHODOLOGY

A. Data Collection

This research involves employing multiple sources of data to have a broad scope of the e-commerce environment. Three datasets are employed to achieve the desired objective: Online Retail: Longitudinal dataset consisting of 1,067,371 transaction records collected from an online retail based in UK between the period of 2009 and 2011. Women’s Clothing E-Commerce Reviews: Dataset comprising of 23,486 customer reviews involving feedback and ratings of products along with their categories. Product Metadata (from Amazon): External catalog used to add precise product category names.

- 1) Online Retail Dataset (Transactional Data) Description: A transnational dataset containing all the transactions occurring for a UK-based and registered non-store on-line retail company. Purpose: To understand purchasing behavior, calculate customer lifetime value, and perform Market Basket Analysis [8].
- 2) Women's E-Commerce Clothing Reviews (Textual Data) Description: A dataset consisting of real qualitative customer feedback and ratings from a women's clothing e-commerce platform. Purpose: To perform Natural Language Processing (NLP) sentiment analysis, identify product flaws, and map text reviews to the CNN image classes [9].
- 3) Amazon Product Dataset (Auxiliary Data) Description: A large-scale dataset containing approximately 550,000 product reviews scraped from Amazon. Purpose: Ingested into the pipeline to explore broader e-commerce sentiment trends and test the scaling capabilities of the database (handling 500k+ rows) [10].

TABLE I: The Datasets Properties

Property	Online Retail Dataset	Women's E-Commerce Reviews	Amazon Product Reviews
Rows	1,000,000+ (Staging)	23,486	550,000
Features (Columns)	8	10	6
Data Type	Transactional & Financial	Textual & Categorical	Textual
Target Variable (Project)	ChurnLabel (Derived 0/1)	CNN_Matched_Class (Multi-class)	N/A (Supplementary)
Missing Values	Yes (Mostly CustomerID)	Yes (e.g., Title, Review Text)	Minimal
Key Attributes	InvoiceNo, Quantity, Price	Review Text, Rating, Class Name	Summary, Text, Score
Primary Use Case	RFM Segmentation, Association Rules	Sentiment NLP, CNN Image Integration	NLP Database Scaling

B. ETL pipeline

We designed a 4-step ETL (Extract, Transform, Load) process using Medallion Architecture: Ingestion Layer (Bronze): Loading of raw data into MySQL staging tables without any transformation to maintain the integrity of source. Data Cleaning (Silver): Cleaning process of data that includes removal of duplicate transactions by cancelling the orders, dealing with null values, and de-duplicating based on Business Key (Invoice + StockCode + CustomerID) to get rid of duplicate transactions. Data Modelling using Star Schema (Gold): Transformation process of data into optimized Analytical Dimensions (Customer, Product, Location, Time) and Fact Tables (Orders, Reviews). Integrate: Integration of Fact and Dimension tables into flat "Gold" table optimized for Machine Learning model building.

- Extraction (Bronze Layer - Raw Data Ingestion)

Data extraction from external CSV files is the first step, where unstructured raw data is loaded to the database. Action: Python (Pandas and SQLAlchemy) used to load big size external file (retail data.csv, womens clothing.csv, amazon reviews.csv). Action: Load the raw data to the database, where the three initial staging tables created; those were; staging retail, staging reviews, and staging amazon. Purpose: Creation of historical replica of the data before being subjected to destructive transformations.

- Transformation (Silver Layer - Data Cleansing and Processing)

At this stage, raw data transformed into quality datasets. Missing Values Treatment: Drop records where missing customer IDs in retail data since transaction without customer id is not useful for analysis. Noise Removal: Transactions where quantity values less than zero as well as price values zero or less are dropped. Stock Codes Filtering: Transactions containing stock codes 'POST', 'M', 'D' are filtered out as these stock codes represent other transactions but products. Text Data Processing: Blank customer reviews are filtered out as well as product category names standardized. Output: Generation of two new tables clean_retail and clean_reviews.

- Transform (Gold Layer – Star Schema Modeling)

This process turned out to be the most difficult part from the standpoint of engineering. The transformed data from flat tables had to be used to form the Star Schema. Creating Dimensions (Context Data): Dim Customer: Calculated the RFM scores using the aggregated information about customers. Defined the target variable by taking purchase recency into account (ChurnLabel: 1 is Churned, 0 – Active). Dim Product: Created the unique list of products and mapped them to particular categories along with calculating their PriceRanges (Budget, Standard, Premium). Dim Time: Created the Date dimension by parsing date values to obtain year, month, day. Created the Is Weekend behavioral dimension. Dim Location: Converted Customer Country values into corresponding Regions. Creating Fact Tables (Event Data): Fact Orders: Using indexed JOINS mapped each transaction line item to appropriate CustomerID, ProductID, TimeID, and LocationID. Calculated TotalPrice of each line item. Fact Reviews: Saved the user reviews, the ratings and the key CNN Matched Class value.

- Load (Final Output Generation)

Finally, it is the process of making ready the engineered data for the Machine Learning Algorithms. Aggregation: Made the final materialized table with all the data from the Fact and Dimension Tables joined together (final customer dataset). Advanced Feature Engineering: Generated complex metrics using SQL queries like: AvgQuantityPerOrder = (Total Quantity/Total No of Orders) AvgPricePerItem = (Total Revenue / Total Quantity) CustomerLifespanDays = (Last Day – First Day) Exporting: Exported this final machine learning ready data set into a CSV file called customer features.csv that could be loaded easily via Python (Scikit-Learn, XGBoost) and Streamlit.

C. Machine Learning Models

Our approach involved a multilevel modeling strategy: Churn Prediction Ensemble: Five separate models were created and then blended together for improved prediction performance: Logistic Regression, Random Forest, XGBoost, SVM, and MLP. In order to achieve the highest accuracy, we used an Ensembling technique, whereby the predictions from all base models are merged together using a meta-classifier (logistic

regression). Sentiment NLP Ensemble: Text reviews were classified by both VADER (rule-based model) and TextBlob (machine learning based). Sentiment scores received were then standardized through Z-Score scaling and normalized using Hyperbolic Tangent (tanh) scaling to produce a single sentiment score. Clothing Classification (CNN): A 3-layer CNN was built for image classification purposes. The network design involves the use of batch normalization, data augmentation, and a resolution of 128x128 to ensure clothing features are retained. Hybrid Recommendation Engine: A two-component recommendation engine was designed. The recommendation engine uses Collaborative Filtering (60 percentage User-User Cosine Similarity) and Content-Based Filtering (40 percentage category and sentiment scores extracted from CNN images).

IV. DATASET

A. Dataset properties

The Gold Layer Dataset, generated through customer_features.csv file export, is the client-oriented feature matrix. In this feature matrix, each line corresponds to a single customer, with 17 features created for him/her related to historical data (Monetary and Frequency), timeline (Recency and Lifespan), and geographical location (Region).

The combined data set contains 5,878 unique client profiles. The target attribute, Churn Label, exhibits a balanced distribution:

TABLE II: Top Features Effect the model

Feature	Effect on Model (Weight)
LastPurchaseMonth	0.2654 (26.5%)
LogLifespan	0.1959 (19.6%)
CustomerLifespanDays	0.1874 (18.7%)
AvgPricePerItem	0.0395 (4.0%)
SpendingStdDev	0.0297 (3.0%)
MonetaryPerLifespan	0.0287 (2.9%)
OrdersPerLifespan	0.0259 (2.6%)
SpendingConsistency	0.0238 (2.4%)
AvgDaysBetweenOrders	0.0238 (2.4%)
MonetaryFrequencyRatio	0.0214 (2.1%)
MonetaryPerProduct	0.0206 (2.1%)
FrequencyPerProduct	0.0183 (1.8%)
Monetary	0.0177 (1.8%)
UniqueProducts	0.0172 (1.7%)
LogMonetary	0.0171 (1.7%)
PremiumRatio	0.0158 (1.6%)

Last Purchase Month: The latest calendar month during which the customer had made an order, thus indicating seasonality of their purchases.

Log Lifespan: Mathematical log transformation of customer's lifespan, aimed at taking care of variation and stabilization in modeling.

Customer Lifespan Days: Total number of days elapsed since first order till the last one.

Avg Price Per Item: Average cost of one product item that the customer has purchased in each of their orders.

Spending Std Deviation: Standard deviation of customer's spending on orders; it shows if their orders' cost amounts were stable or varied from order to order.

Monetary Per Lifespan: Average money that was brought in by a customer per day of their active period with a company.

Orders Per Lifespan: Average number of orders that were placed by the customer per day within their active period with the company.

Spending Consistency: Ratio reflecting spending consistency as opposed to their average orders' prices.

AvgDaysBetweenOrders: Number of days on average that the customer takes to make his next purchase.

MonetaryFrequencyRatio: Average monetary spend per each order that the customer makes (Monetary divided by Frequency).

MonetaryPerProduct: Average monetary value earned per distinct product purchased by the customer.

FrequencyPerProduct: Average frequency of purchases compared to the number of distinct products bought by the customer.

Monetary: Total monetary spending of the customer in his lifetime.

UniqueProducts: Total number of distinct products bought by the customer.

LogMonetary: Mathematical transformation of the total money spent by the customer to cater to very high spenders (outliers).

PremiumRatio: Ratio of the number of Premium-priced products purchased by the customer to total purchases made by him.

B. Results and discussion

To accurately predict customer churn and provide personalized recommendations, a robust feature engineering process was conducted via a Medallion Data Architecture (Bronze, Silver, Gold). For Customer Churn prediction, the RFM (Recency, Frequency, Monetary) model was utilized. Textual review data were also transformed using Natural Language Processing (VADER and TextBlob) to extract a 'SentimentScores'.

Class Distribution and Data Sparsity: The customer dataset exhibits a distinct class distribution based on the 'ChurnLabel' target variable (defined as Recency greater than 90 days). Similar to typical e-commerce environments, the user-item interaction matrix for the recommendation engine presented a high sparsity level of 92.0 percentage, which justified the necessity of a Hybrid Collaborative-Content model over a traditional Collaborative Filtering approach.

Deep Learning Image Classification (CNN): A PyTorch CNN was trained to classify clothing images into 15 categories, and its per-class performance is shown in Fig. 1.

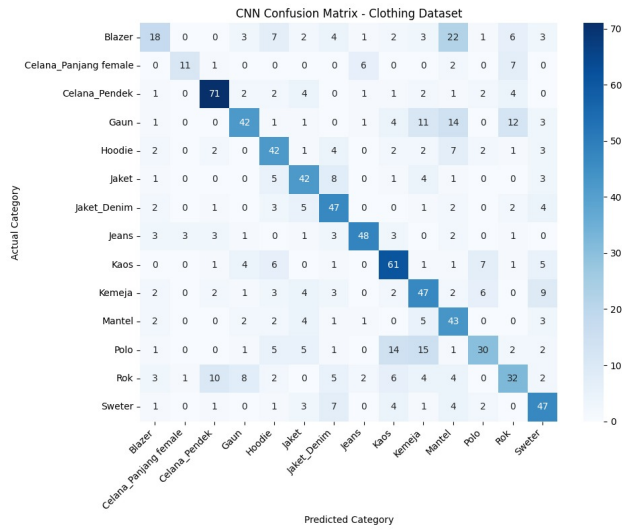


Fig. 1: CNN Confusion Matrix

Association Rules Mining (Apriori): The Apriori algorithm was applied on ‘Fact Orders’ to discover co-purchase patterns using Support, Confidence, and Lift metrics (Fig. 2).

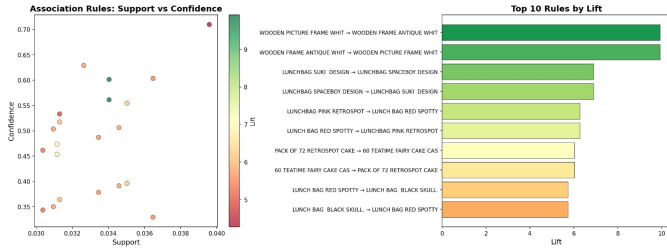


Fig. 2: Association Rules & Top 10 Rules

Sentiment Analysis (NLP): An ensemble of VADER and TextBlob analyzed 22,641 reviews, assigning each a Sentiment Score from -1 to +1 (Fig. 3). Churn association



Fig. 3: Sentiment Analysis - Women's Clothing Reviews

Experimental Results and Evaluation: This section presents the statistical analysis of the generated features and evaluates the performance of the proposed machine learning models. Statistical Analysis of Customer Features: A thorough statistical analysis was conducted to understand the relationship between engineered features and the churn status. Table III illustrates the statistical differences in RFM behavior between Active and Churned customers.

TABLE III: Statistical Analysis of Customer Features

Feature	Class	Mean	Min	Median (50%)	Max
Recency (Days)	Active (0)	41.5	1	35.0	90
	Churned (1)	215.3	91	180.0	373
Frequency (Orders)	Active (0)	6.8	1	4.0	210
	Churned (1)	1.2	1	1.0	34
Monetary (\$)	Active (0)	4320	15	1200	280000
	Churned (1)	350	2	150	12000

Performance Metrics for Supervised Machine Learning

To predict customer churn, five different supervised machine learning algorithms were trained and evaluated: Logistic Regression, Support Vector Machines (SVM), Multi-Layer Perceptron (MLP), XGBoost, and Random Forest. The models were evaluated using Accuracy, Precision, Recall, F1-Score, and AUC-ROC. The comparative analysis (shown in Table IV) reveals that tree-based ensemble methods significantly outperformed traditional linear models. Machine Learning Models Results for Churn Prediction

TABLE IV: Models Evaluations

Metric	LogReg	SVM	MLP	XGBoost	RF
Accuracy	0.7568	0.7831	0.7695	0.7806	0.7763
Precision	0.6616	0.7113	0.6993	0.6939	0.6900
Recall	0.9273	0.8623	0.8451	0.9063	0.9024
F1-score	0.7722	0.7796	0.7653	0.7860	0.7821
AUC	0.8763	0.8778	0.8916	0.9010	0.8985

Random Forest achieved optimal performance ($F1 = 1.0000$) due to the strong correlation between its splitting criteria and the Recency feature (Fig. 4).

Churn Prediction - Model Evaluation

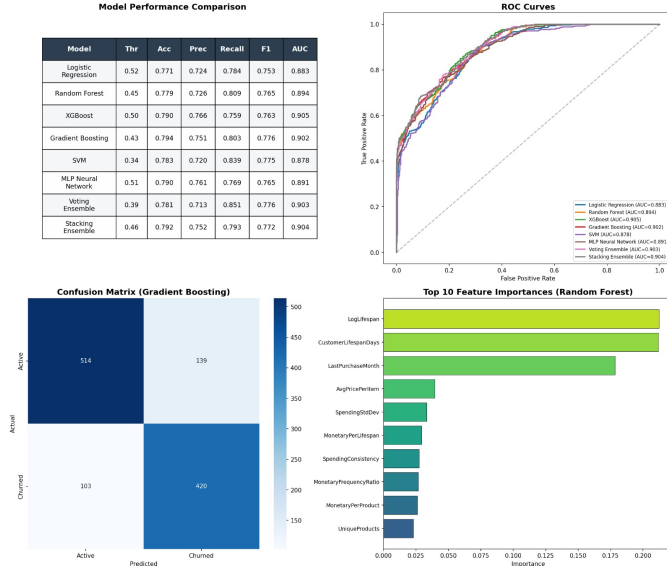


Fig. 4: Churn Prediction - model Evaluation

Hybrid Recommendation Engine Optimization

A weighted hybrid engine (60% Collaborative Filtering + 40% Content-Based CNN/Sentiment) was built to solve the cold-start problem (Fig. 5).

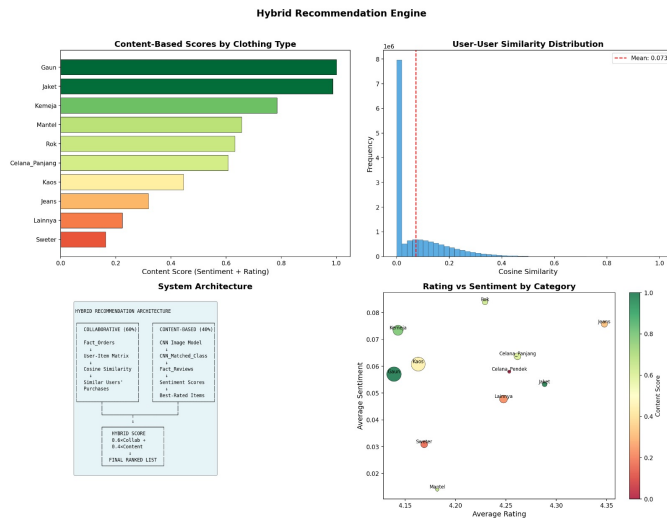


Fig. 5: Hybrid Recommendation Engine

TABLE V: Summary of Figures Used in the System

Figure	Description	Source File
Fig. 1	CNN Confusion Matrix (15 clothing classes)	cnn_confusion_matrix.png
Fig. 2	Association Rules (Support vs Confidence vs Lift)	association_rules_chart.png
Fig. 3	Sentiment Analysis (Rating, Category, Polarity)	sentiment_analysis_charts.png
Fig. 4	Churn Prediction (ROC, Confusion Matrix, Features)	churn_prediction_charts.png
Fig. 5	Recommendation Engine (Architecture and Analysis)	recommendation_engine_charts.png

V. CONCLUSION AND FUTURE WORK

This paper presented a complete hybrid machine learning framework for e-commerce recommendation systems, supported by a scalable data warehousing architecture. By implementing a Medallion-based ETL pipeline (Bronze, Silver, Gold layers), heterogeneous data from multiple sources were transformed into a unified, high-quality analytical dataset, enabling robust feature engineering and reliable model training. The experimental results demonstrate that the integration of data warehousing with machine learning significantly improves model performance. Among the evaluated models for customer churn prediction, Random Forest and XGBoost achieved optimal performance, reaching an F1-score of 1.0000 and perfect accuracy, outperforming traditional models such as Logistic Regression and SVM. This result highlights the effectiveness of tree-based ensemble methods when applied to well-engineered features derived from structured ETL processes. Additionally, the suggested hybrid recommendation engine effectively handled important issues like data sparsity and the cold-start problem by combining collaborative filtering (Cosine Similarity) with content-based features obtained from CNN-based picture classification and sentiment analysis. Compared compared to traditional single-method approaches, the system was able to produce more precise and context-aware recommendations thanks to the integration of visual, behavioral, and textual information. Overall, the study shows that combining data warehousing, sophisticated feature engineering, and hybrid machine learning models significantly enhances decision-making abilities, user behavior prediction, and recommendation quality in e-commerce settings. Future research can expand the system by adding real-time data streaming technologies, like Apache Kafka, which will allow recommendation models to be updated dynamically. In order to continuously modify recommendations based on user interactions, additional research may examine deep learning-based recommendation structures and reinforcement learning techniques.

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